

MC Sachs two-point: numerical quantities, definitions, and comparison with analysis-3

analyses/mc_sachs_2pt/ (internal reference)

2026-06-06

Every quantity is given with its exact mathematical definition and the code symbol that implements it. Lengths are physical Mpc; the affine parameter λ is 0 at the observer and increases toward the source λ_f . Component convention: $s = (s_0, s_1, s_2) = (\Phi_{00}, \text{Re } \Psi_0, \text{Im } \Psi_0) = (\kappa\text{-rate}, \gamma_+\text{-rate}, \gamma_\times\text{-rate})$. A two-ray bundle uses $\hat{n}_1, \hat{n}_2$ with $\hat{n}_1 \cdot \hat{n}_2 = \cos \gamma$; the 6-vector is $f = (f(\hat{n}_1), f(\hat{n}_2))$, indices 0–2 = ray 1, 3–5 = ray 2.

1 Dynamics and the observable

Stochastic Sachs fluctuation ODE (per ray; F is the local quadratic vertex):

$$\frac{ds_a}{d\lambda} = -2\theta_{sa}(\lambda) s_a + F_{abc} s_b s_c + f_a(\hat{n}, \lambda). \quad (1)$$

Convergence/shear observable (the `integrate_over='all'` observable):

$$\kappa_a(\hat{n}) = \int_0^{\lambda_f} s_a(\hat{n}, \lambda) d\lambda. \quad (2)$$

F-vertex (paper Table 1 = `inputs/F_tensor.npy`; nonzero entries)

$$F_{000} = -1, \quad F_{011} = -1, \quad F_{022} = -1, \quad F_{101} = -2, \quad F_{202} = -2, \quad (3)$$

so $F_{abc} s_b s_c = (-(s_0^2 + s_1^2 + s_2^2), -2s_0 s_1, -2s_0 s_2)$ (`_F, _f_vertex`). The 2-point observable is the cross-ray correlation $\xi_{ab}(\gamma) = \langle \kappa_a(\hat{n}_1) \kappa_b(\hat{n}_2) \rangle$; every estimator returns this (3,3) cross-ray block.

2 Background quantities (background.py over D_callable.py)

Quantity	Definition	Code
Jacobi amplitude	$D(\lambda) = a(\lambda) \chi(\lambda)$ (flat FLRW closed form)	<code>Background.D</code>
Focusing rate	$\theta_{sa}(\lambda) = \frac{d \ln D}{d\lambda} = D'/D$	<code>Background.theta_sa</code>
Linear response	$R(t, \lambda') = \Theta(t - \lambda') [D(\lambda')/D(t)]^2$	<code>Background.response_R</code>
LOS window	$W(\lambda) = \int_\lambda^{\lambda_f} \left[\frac{D(\lambda')}{D(t)} \right]^2 dt = \int_\lambda^{\lambda_f} R(t, \lambda) dt$	<code>driver_stats.order0_window</code>

R is the exact Green's function of the linear ($F=0$) part of (1): $\partial_t R = -2\theta_{sa}(t)R$, $R(\lambda', \lambda') = 1$. The linear solution is $s_a^{(1)}(\hat{n}, \lambda) = \int_0^\lambda R(\lambda, \lambda') f_a(\hat{n}, \lambda') d\lambda'$, so from (2) $\kappa_a^{(1)}(\hat{n}) = \int_0^{\lambda_f} W(\lambda') f_a(\hat{n}, \lambda') d\lambda'$: W is the window with which the observable sees a driving-field impulse. Cosmology (hardcoded in `D_callable`): flat Λ CDM, $\Omega_m = 0.31609$, $h = 0.6711$, $z_s = 5 \Rightarrow \lambda_f = 2313.029$ (analysis-3 baseline).

3 Bare driving-field cumulants (driver_stats.py)

The driving field is white in λ with angular cumulant densities.

3.1 2-cumulant Σ^2 (Sigma2Builder, sigma2_matrix)

$$\langle f_a(\hat{n}_1, \lambda) f_b(\hat{n}_2, \lambda') \rangle_c = \Sigma_{ab}^2(\hat{n}_1 \cdot \hat{n}_2; \lambda) \delta(\lambda - \lambda'). \quad (4)$$

Σ^2 is obtained by *de-windowing* the corr-op propagator $C_{ab}(\cos; \lambda_1, \lambda_2) = \langle \Phi_a(\hat{n}_1, \lambda_1) \Phi_b(\hat{n}_2, \lambda_2) \rangle$ (the R -windowed 2-cumulant). Since $R = [D/D]^2$ is scalar,

$$\Sigma_{ab}^2(\cos; \lambda) = D(\lambda)^{-4} \frac{d}{d\lambda} [D(\lambda)^4 C_{ab}(\cos; \lambda, \lambda)], \quad (5)$$

the exact inverse of $C_{ab}(\cos; t, t) = \int_0^t [D(\ell)/D(t)]^4 \Sigma_{ab}^2(\cos; \ell) d\ell$. `Sigma2Builder.matrix` splines $D^4 C_{ab}(t, t)$ and differentiates; the optional anchor $C_0 = 1/0.881 = 1.135$ (`apply_c0`) rescales Σ^2 so the white-noise Order-0 matches analysis-3 in absolute normalization. The two-ray (6,6) matrix `sigma2_6x6` uses within-ray ($\cos=1$) diagonal blocks and cross-ray ($\cos = \cos \gamma$) off-diagonal blocks.

3.2 Order-0 analytic prediction O_0 (order0_mc)

The linear ($F=0$) limit of $\langle \kappa \kappa \rangle$; (4) collapses the double LOS integral to one:

$$O_0^{ab}(\gamma) = \int_{\lambda_{\min}}^{\lambda_f} \Sigma_{ab}^2(\cos \gamma; \lambda) W(\lambda)^2 d\lambda, \quad \lambda_{\min} = 406. \quad (6)$$

This is the MC's *exact* linear target.

3.3 3-cumulant ζ (zeta_tensor, zeta6)

Equal-time third cumulant (delta-correlated in λ):

$$\langle f_a(\hat{n}_1, \lambda) f_b(\hat{n}_2, \lambda') f_c(\hat{n}_3, \lambda'') \rangle_c = \zeta_{abc}(\hat{n}_1, \hat{n}_2, \hat{n}_3; \lambda) \delta(\lambda - \lambda') \delta(\lambda - \lambda''). \quad (7)$$

`zeta_tensor` wraps the project `equal_time_limber` $\kappa^{(3)}$ vertex: a symmetric (3,3,3) tensor indexed by the cosine-triple ($\hat{n}_i \cdot \hat{n}_j$) and λ (non- R -contracted, lambda-density).

3.4 Skewness deformation Q (solve_Q)

For a node covariance M^2 and a target node 3-cumulant Z , solve

$$Z_{abc} = Q_{amn} M_{mb}^2 M_{nc}^2 + Q_{bmn} M_{ma}^2 M_{nc}^2 + Q_{cmn} M_{ma}^2 M_{nb}^2, \quad (8)$$

for Q symmetric in its last two indices (reduces to the scalar $Q = z/(3m^2)$).

4 Monte-Carlo engine (sachs_mc_core.py)

4.1 Exact-propagator discretization

Uniform grid $\lambda_k = \lambda_{\min} + k d\lambda$, $d\lambda = (\lambda_f - \lambda_{\min})/(N-1)$, $\lambda_{\min} = 406$. Response applied *into* node k , then drift + noise injected with unit response:

$$\text{resp}[k] = [D(\lambda_{k-1})/D(\lambda_k)]^2, \quad \text{resp}[0] = 1, \quad (9)$$

$$s_k = \text{resp}[k] s_{k-1} + (F_{abc} s_b s_c)|_{k-1} d\lambda + (\text{noise})_k, \quad (10)$$

$$\kappa = \sum_k w_k s_k, \quad w_k = d\lambda \left(\frac{1}{2} d\lambda \text{ at } k=0, N-1 \right) [\text{trapezoid}]. \quad (11)$$

(`_step`, `_precompute`, `_field_precompute`). With $F=0$, (10) telescopes to $s_k = \sum_{j \leq k} [D(\lambda_j)/D(\lambda_k)]^2 dW_j$, so the Gaussian MC reproduces (6) by construction.

4.2 Noise models

White (`_precompute`; used by `simulate`, `simulate_ff_crn`): the increment dW_k is Gaussian with $\text{Cov}(dW_k) = \Sigma_{6 \times 6}^2(\cos \gamma; \lambda_k) d\lambda$.

Colored AR(1) (`_field_precompute`; used by the FK estimators): a field with stationary node covariance and correlation length σ_λ ,

$$V_k = \frac{\Sigma_{6 \times 6}^2(\cos \gamma; \lambda_k)}{2\sigma_\lambda}, \quad \rho = e^{-d\lambda/\sigma_\lambda}, \quad z_k = \rho z_{k-1} + \sqrt{1 - \rho^2} L_k \xi_k, \quad L_k L_k^\top = V_k, \quad (12)$$

injected as $(\text{noise})_k = f_k d\lambda$. The factor $1/(2\sigma_\lambda)$ enforces $\int \langle z(\lambda) z(\lambda') \rangle d\Delta = 2\sigma_\lambda V = \Sigma^2$ (one-delta collapse), recovering the white density (4) as $\sigma_\lambda \rightarrow 0$.

Skew injection (`skew_scale` $\neq 0$): deform by a local quadratic,

$$f_{s,k} = z_k + \frac{1}{2} Q_k (z_k \otimes z_k - V_k), \quad Q_k = \text{skew} \cdot \text{solve_Q} \left(V_k, \frac{\zeta_{6,k}}{(2\sigma_\lambda)^2} \right). \quad (13)$$

The node target $\mu_{3,k} = \zeta_{6,k}/(2\sigma_\lambda)^2$ (two-delta collapse) gives $\iint \langle f_s f_s f_s \rangle d\Delta_1 d\Delta_2 = \zeta_{6,k}$, reproducing (7). Note $Q_k \sim \zeta_6/(3(\Sigma^2)^2) \sim -1.4 \times 10^7$ is σ_λ -independent; the field skewness $\sim \zeta_{6,\text{node}}/V^{3/2} \sim 4-6$ grows as $\sigma_\lambda^{-1/2}$ (Lévy).

5 Channels and estimators

The MC yields only *full* (all-orders-in- F) correlations, each compared to the sft-wick partial sum truncated at order 2.

(a) Gaussian full (`simulate`): white, F on, no skew; $\xi = [(1/N_{\text{good}}) \sum \kappa \otimes \kappa]_{[0:3, 3:6]}$. Contains $O_0 + \text{FF} + \dots$; `o0_analytic`=(6).

(b) FF via F -toggle CRN (`simulate_ff_crn`): two arms share the *same* white noise, F on vs off, differenced per realization:

$$\text{FF}_{ab}^{\text{mom}} = \langle \kappa_a \kappa_b \rangle_{F\text{-on}} - \langle \kappa_a \kappa_b \rangle_{F\text{-off}}, \quad (14)$$

$$\text{FF}_{ab}^{\text{disc}} = \langle \kappa_a \rangle_{F\text{-on}} \langle \kappa_b \rangle_{F\text{-on}} - \langle \kappa_a \rangle_{F\text{-off}} \langle \kappa_b \rangle_{F\text{-off}}, \quad (15)$$

$$\text{FF}_{ab}^{\text{conn}} = \text{FF}_{ab}^{\text{mom}} - \text{FF}_{ab}^{\text{disc}}. \quad (16)$$

Since $\langle \kappa \rangle_{F\text{-off}} = 0$, $\text{FF}^{\text{disc}} = \langle \kappa \rangle_2 \otimes \langle \kappa \rangle_2$, the square of the mean 2nd-order convergence $\langle \kappa \rangle_2 = \int W F_{abc} \langle s_b s_c \rangle$.

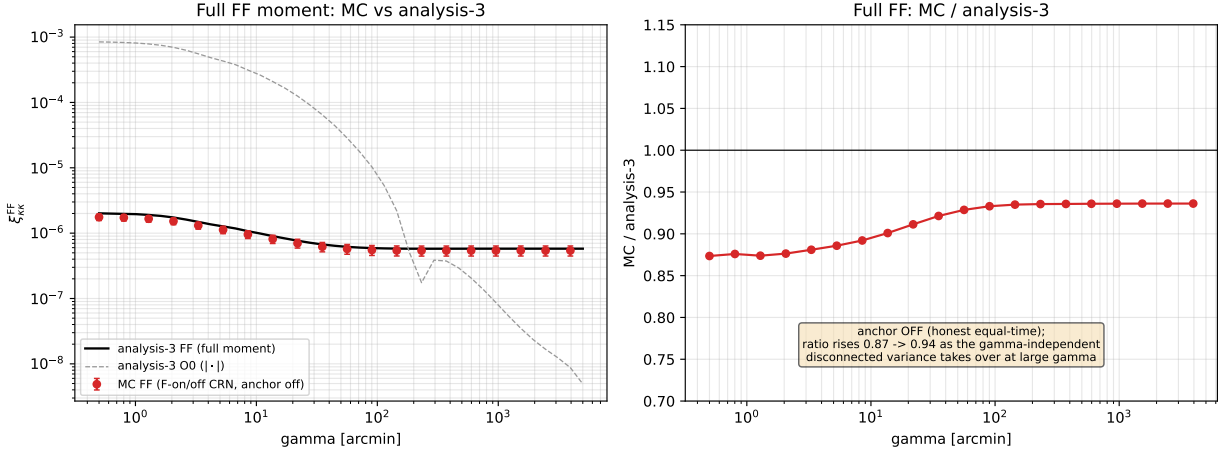


Figure 1: FF channel (Task 1): the FULL FF moment, MC vs analysis-3 (no connected/disconnected split). *Left*: analysis-3 FF (line) vs the MC F -on/off CRN moment (markers, anchor off), with O0 for context. *Right*: the ratio rises from ~ 0.87 (small γ) to ~ 0.94 (large γ) – the equal-time (white-noise) approximation of the 2-cumulant, the same finite- λ -correlation behind the O0 anchor 0.881, larger at small γ (where $\text{FF} \sim (\Sigma^2)^2$) and smaller at large γ (where the γ -independent, exactly-de-windowed diagonal variance dominates).

(c) **Analytic equal-time FK** (`fk_analytic.py`). The order-2 FK diagram (one F -vertex, one ζ) has four R -propagators and zero C : $\text{obs}_a \xrightarrow{R} F \xrightarrow{R,R} \text{two } \zeta \text{ legs}$; third ζ leg $\xrightarrow{R} \text{obs}_b$. The two F -side legs are co-located on $\hat{n}_a$ (triple $(1, \cos \gamma, \cos \gamma)$); the equal-time δ of (7) collapses the 4-D time integral to 2-D, and `integrate_over='all'` turns each external R into W :

$$\xi_{ab}^{\text{FK}}(\gamma) = \int_0^{\lambda_f} d\tau \int_0^\tau d\lambda_v W(\tau) W(\lambda_v) \left[\frac{D(\lambda_v)}{D(\tau)} \right]^4 G_{ab}(\lambda_v; \gamma) \quad (17)$$

$$G_{ab} = (M + M^\top)_{ab}, \quad M_{ab} = \sum_{m,n} F_{amn} Z_{bmn}, \quad Z = \zeta(\hat{n}_1, \hat{n}_1, \hat{n}_2; \lambda_v). \quad (18)$$

The external index a rides the F -vertex (F_{amn}), only b rides one ζ leg (Z_{bmn}); $M + M^\top$ is the $\hat{n}_1 \leftrightarrow \hat{n}_2$ symmetrization; the MSR/combinatorial prefactor is 1 (calibrated on the validated demo2 integrator). For $\kappa\kappa$: $G_{00} = -2(\zeta_{\text{TTT}} + \zeta_{\text{Bmod}})$. Code reduces (17) to two 1-D quadratures via $u = D/D(\lambda_f)$: $\xi_{ab}^{\text{FK}} = \int d\lambda_v W u^4 G_{ab} H$, $H(\lambda_v) = \int_{\lambda_v}^{\lambda_f} W(\tau) u(\tau)^{-4} d\tau$.

(d) **FK Monte-Carlo, variance-reduced** (`simulate_fk_vr`). FK is linear in Q , and obs_b connects to its ζ leg F -free, so the no- F windowed response $\kappa_{d,B}^{(0)} = \int W \delta f_B$ suffices (no s_d ODE). Pull Q *outside* the average:

$$T_{AB} = \frac{1}{2} d\lambda \sum_k W(\lambda_k) \sum_{ij} Q_k[B, i, j] C_{A,ij}(k), \quad C_{A,ij}(k) = \text{Cov}(\kappa_{g,A}^F, z_i(\lambda_k) z_j(\lambda_k)), \quad (19)$$

$$\text{FK}_{ab} = (T + T^\top)_{[a, 3+b]} \quad (\text{cross-ray block}). \quad (20)$$

Three ingredients (all required): (i) Q_k multiplies the *averaged* C , never the per-sample $w_{ij} = z_i z_j - V$; (ii) C is the *matched sample* covariance $\langle \kappa_q z_i z_j \rangle_{\text{batch}} - \langle \kappa_q \rangle \langle z_i z_j \rangle_{\text{batch}}$ (the disconnected

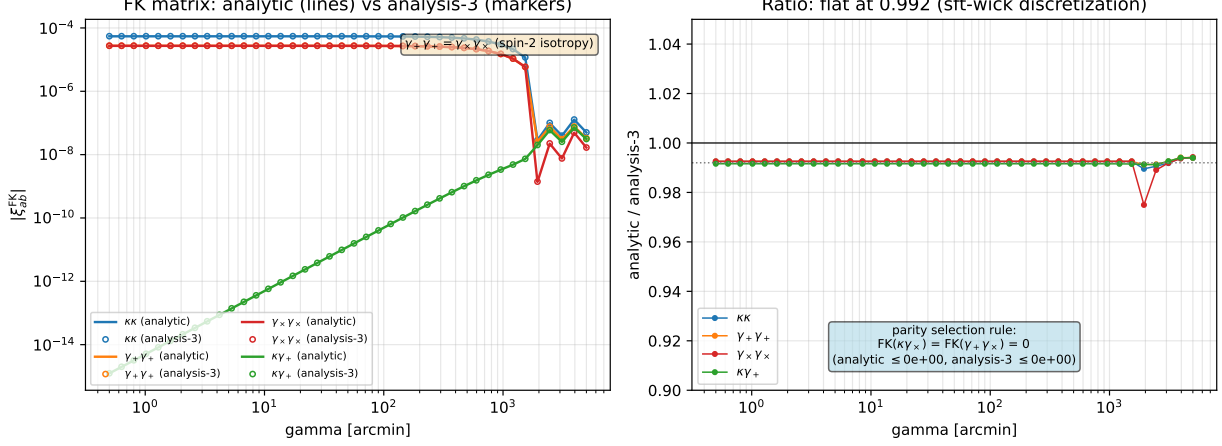


Figure 2: Analytic FK (17) vs analysis-3 over the FULL (3,3) component matrix. *Left*: $|\xi_{ab}^{FK}|$ vs γ – analytic lines, analysis-3 markers; $\gamma_+\gamma_+ = \gamma_+\gamma_-$ (spin-2 isotropy). *Right*: the ratio is flat at 0.992 for every nonzero pair (the 0.8% is sft-wick’s coarser internal discretization), and the parity selection rule $\xi^{FK}(\kappa\gamma_+) = \xi^{FK}(\gamma_+\gamma_-) = 0$ holds exactly in both.

piece is $\sim 10^7 \times$ the signal, so it must be removed with the same-sample $\langle z_i z_j \rangle$, *not* the theoretical V_k ; (iii) $\kappa_g^F = \kappa_g(F\text{-on}) - \kappa_g(F\text{-off})$ (shared-noise CRN) drops the linear L_1 noise $\sim 1000 \times$ and fixes the sign. The naive equivalent `simulate_fk_pathA` is blow-up-free but statistics-hungry; kept for reference only.

6 Comparison with analysis-3

Analysis-3 npz under `sftwick_outputs/2PCF/`: dirs `C_corr_op_00`, `C_corr_op_K_limber_FF`, `C_corr_op_K_limber_FK`. The $\kappa\kappa$ channel = rows `a==0` & `b==0`; $\gamma = \deg(\arccos \hat{x} \cdot \hat{y}) \cdot 60$; $\lambda_f = 2313.029$. All inputs (Σ^2, ζ, F, R , observable) are the *same* objects the MC uses.

The analysis-3 tags below abbreviate the three npz directories named above.

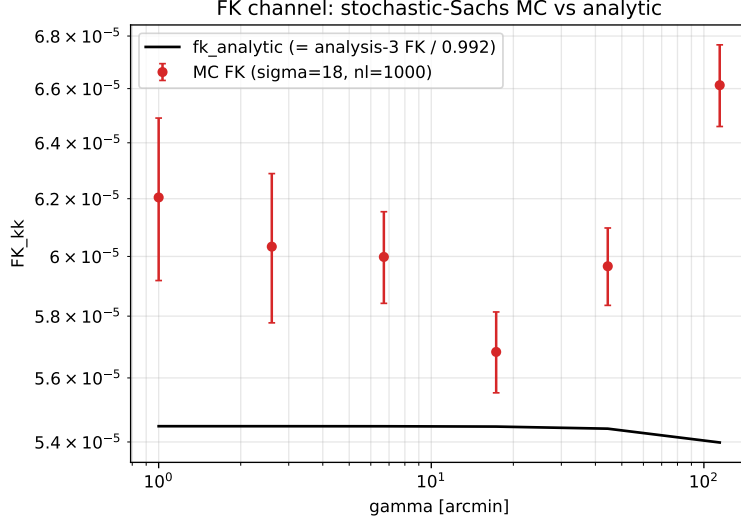


Figure 3: FK Monte-Carlo (19) (variance-reduced, $\sigma_\lambda=18$, $n_\lambda=1000$) vs the analytic equal-time FK over γ . The stochastic-Sachs MC reproduces the analytic FK to ~ 10 –25% (ratio ~ 1.04 –1.23, flat), with correct sign – an independent confirmation of the 3-cumulant channel.

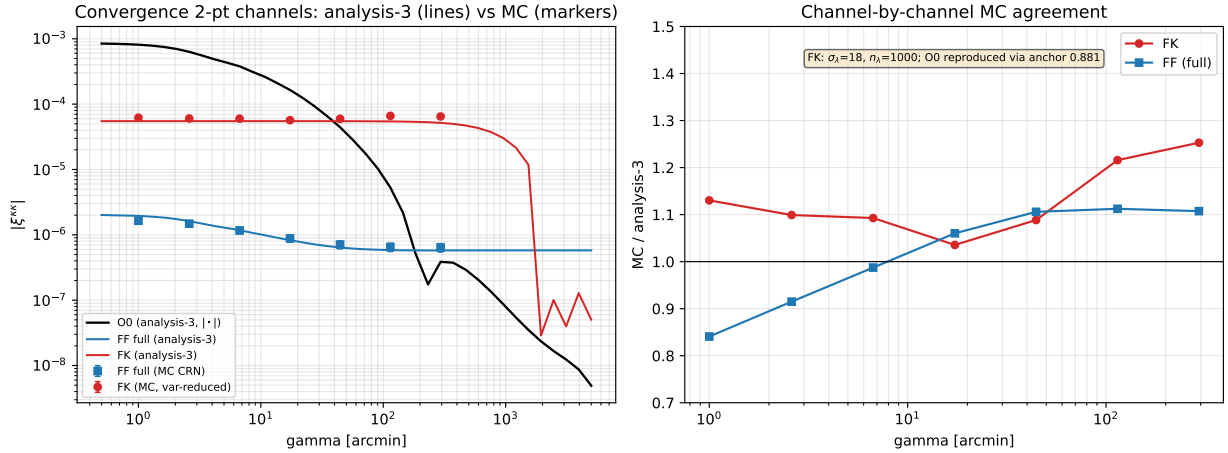


Figure 4: The convergence two-point function $\xi^{\kappa\kappa}(\gamma)$ by channel: O0 / FF (full moment) / FK – analysis-3 (lines) vs Monte-Carlo (markers). *Left*: O0 dominates and crosses zero ($\sim 200'$); FK $\sim 5 \times 10^{-5}$ is nearly flat; the full FF decays toward its γ -independent large- γ value. *Right*: channel-by-channel MC/analysis-3. FK ~ 1.1 (var-reduced estimator); FF (full) ~ 0.9 (the equal-time deficit, rising toward 1 at large γ); O0 is reproduced via the anchor 0.881. (Replaces the archived apples/oranges F -off overlay.)

Channel	MC quantity	analysis-3	Result
Order 0	<code>order0_mc</code> , (6)	00	ratio = 0.881 (γ -indep; $C_0 = 1.135$ absorbs it; the 12% is the equal-time collapse dropping the cross-shell tail).
FF (ord. 2)	<code>simulate_ff_crn</code> , (14)–(16)	FF (limber)	FULL FF moment: MC/analysis-3 ~ 0.87 (small γ) $\rightarrow 0.94$ (large γ), the equal-time deficit (rises as the γ -independent diagonal piece takes over). WHY the floor: analysis-3's flat large- γ value 5.785×10^{-7} is the disconnected $\langle \kappa \rangle_2^2$ – code-confirmed (sft-wick FF = 5 conn +1 <i>disconnected</i> $R(x, y_0)R(y, y_1)C(y_0, y_0)C(y_1, y_1)$; $\langle \kappa \rangle_2^2$ from corr-op C = floor to 0.5%). The de-windowed diagonal variance is reproduced exactly, only the cross-shell ($\sim (\Sigma^2)^2$) is deficient $\Rightarrow$ the γ -rising ratio.
FK (ord. 2)	<code>fk_analytic</code> , (17)	FK (limber)	ratio = 0.992, <i>flat</i> across the full (3,3) matrix (kk peak 5.449 vs 5.489×10^{-5} ; parity-zero pairs (0,2), (1,2) exactly 0; $FK_{11} = FK_{22}$). The 0.8% is sft-wick's coarser internal discretization (stable under refinement).
FK (ord. 2)	<code>simulate_fk_vr</code> , (19)	via analytic	MC/analytic ~ 1.04 – 1.23 (flat over $\gamma = 1'$ – $114'$, SE $\sim$ few %), sign correct. Independent stochastic-ODE confirmation to ~ 10 – 25% . Grid convergence is non-monotonic (fine-grid heavy-tailed upward bias); a clean $< 10\%$ $\sigma_\lambda \rightarrow 0$ limit is future work.

Consistency (doc vs code, numerically verified 2026-06-06).

1. $R(t, \lambda') = [D(\lambda')/D(t)]^2$ (exact).
2. (6) lower-limit window: W vs `scipy.quad`, rel 2×10^{-10} .
3. `order0_mc` = $\int \Sigma_{00}^2 W^2 d\lambda$ (independent quad), rel 2×10^{-4} .
4. `fk_analytic` = brute-force 2-D quadrature of (17), rel 2×10^{-2} (coarse reference grid; confirms the integrand structure).
5. colored field $2\sigma_\lambda V_{00} = \Sigma_{00}^2$, i.e. $\int \langle zz \rangle d\Delta = \Sigma^2$ (exact).

Figures (in `figures/`): `xi_kappa_channels` (full O0/FF/FK decomposition, MC vs analysis-3), `fk_matrix_validation` (FK (3,3) matrix + selection rules), `fk_mc_vs_analytic` (FK MC γ -sweep), `ff_channel_mc_vs_analysis3` (FF decomposition), `anchor_FK_check`, `anchor_00_check` (all .pdf). [The earlier apples/oranges `mc_vs_analysis3_xi_kappa` is archived under `figures/_archive_*`.] Companion notes: `FF_NOTES.md` (Task 1), `FK_NOTES.md` (FK), `DESIGN.md` (architecture).